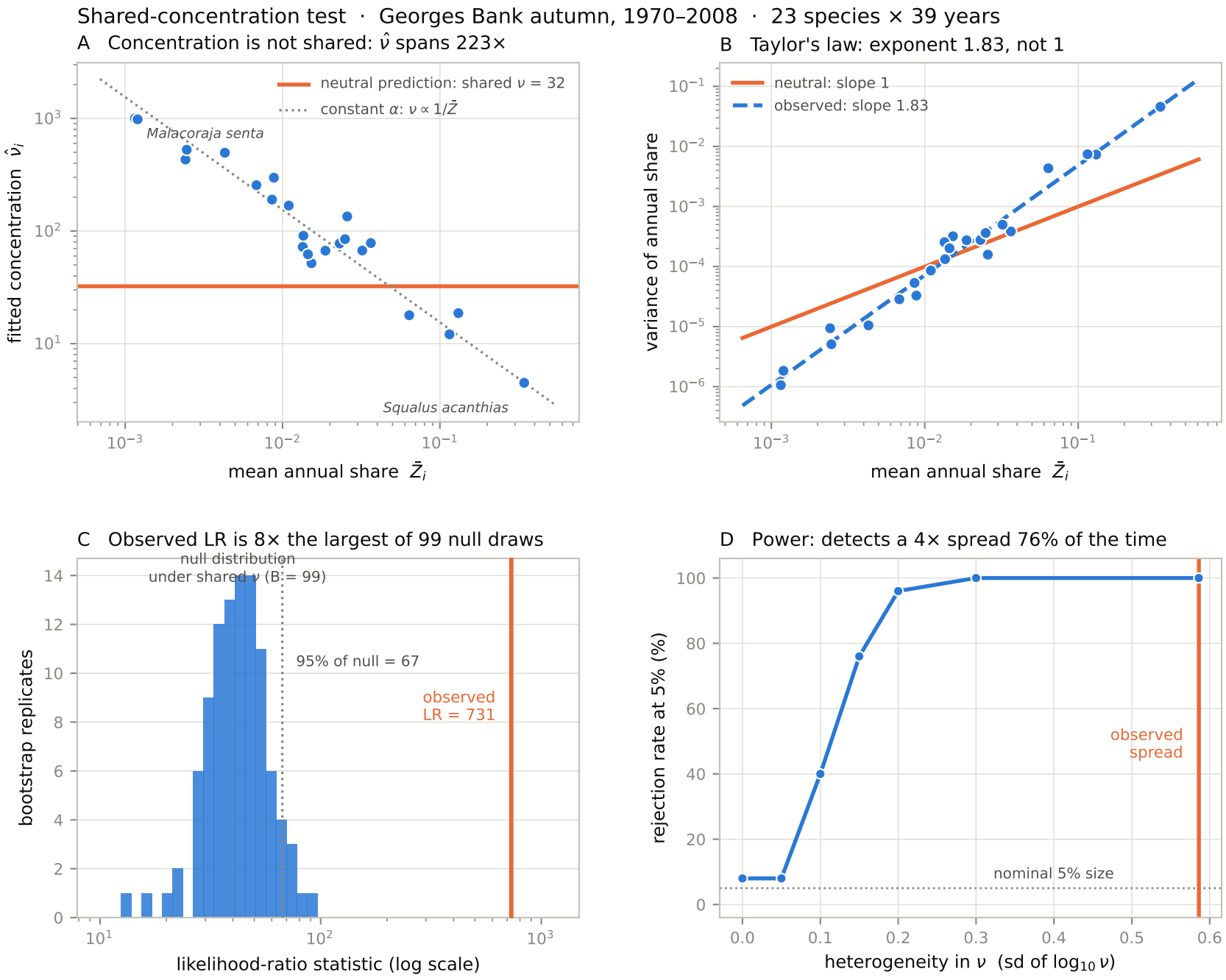
Shared-concentration test · Georges Bank autumn, 1970–2008 · 23 species × 39 years
A Concentration is not shared: $\hat{\nu}$ spans 223×
neutral prediction: shared ν = 32
constant α: $\nu \propto 1/\bar{Z}$
Malacoraja senta
Squalus acanthias
fitted concentration $\hat{\nu}_i$
mean annual share $\bar{Z}_i$
B Taylor's law: exponent 1.83, not 1
neutral: slope 1
observed: slope 1.83
variance of annual share
mean annual share $\bar{Z}_i$
C Observed LR is 8× the largest of 99 null draws
null distribution
under shared ν (B = 99)
95% of null = 67
observed
LR = 731
bootstrap replicates
likelihood-ratio statistic (log scale)
D Power: detects a 4× spread 76% of the time
observed
spread
nominal 5% size
rejection rate at 5% (%)
heterogeneity in ν (sd of $\log_{10}\nu$)